



MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY

Project Proposal

RMG SMV MANAGEMENT SYSTEM

CSE-302: Database Management System Sessional

Group – 02

Section - B

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1. Introduction

RMG SMV MANAGEMENT SYSTEM

This project involves the development of a comprehensive database management system designed to streamline the daily operations of a Ready-Made Garments (RMG) manufacturing unit. The system tracks the complete lifecycle of a garment order, from initial placement by the buyer and raw material procurement, through production and quality inspection, to final shipment and financial settlement. By mapping out the exact relationships between departments, personnel, and resources, the database ensures all operational data remains centralized and accessible.

The operational cycle begins when a Buyer places an Order, which explicitly outlines the manufacturing requirements. This Order dictates the required Material, which the system structurally categorizes into specific subtypes, namely Fabric and Accessories. To fulfill these production requirements, the company relies on a Supplier who provides the necessary Material to begin the manufacturing process.

Once the material requirements are set, the financial and planning phases begin. The Accounts department informs the creation of a Bill of Materials (BOM), detailing the specific items needed for the production batch. The BOM, alongside the operational costs associated with the Machinery used on the factory floor, directly adds to the overall Costing profile. This Costing profile is strictly tied to the specific Order to ensure profit margins and budgets are accurately maintained for each client.

With materials and costs finalized, the workflow shifts to the factory floor, where the workforce is managed under a distinct Employee hierarchy. An Employee is classified as either an Incharge or a Worker. An Incharge manages a specific Production stage, while the Worker actively executes the labor on that same stage. During this phase, the Production stage utilizes the factory's Machinery to physically transform the raw materials into garments.

As garments complete the manufacturing phase, the output from the Production stage goes directly for Inspection to ensure strict quality standards are met. Once the items pass this quality check, the Inspection phase yields the approved Final Product. The Final Product then goes to Packaging to be securely prepped for transport. Once packed, the goods are sent for Shipment, concluding the physical supply chain when the Buyer ultimately receives the Shipment.

Running parallel to this physical supply chain is a robust financial tracking system managed entirely by the Accounts department. Accounts is responsible for managing outgoing capital by paying both the Supplier for raw materials and the Employee for their labor. On the incoming capital side, the initial Order determines the total Payment required from the client. The Buyer makes this Payment, which the Accounts department directly handles and reconciles to successfully close out the operational cycle.

2. Project Objective

The primary objective of the Ready-Made Garments (RMG) Management System is to digitize, centralize, and automate the complex operational and financial workflows of a garment manufacturing unit. By replacing fragmented manual tracking with an integrated Oracle database solution, the system aims to provide real-time visibility into every stage of the supply chain. This includes optimizing raw material procurement, accurately calculating production costs, monitoring floor-level worker output, ensuring quality control, and streamlining financial reconciliations. Ultimately, the project seeks to reduce operational bottlenecks, minimize material waste, and improve overall order turnaround times from buyer initiation to final shipment.

3. Users

The system is designed to be utilized by several key stakeholders within the manufacturing organization, each with specific role-based access. **Accounts Managers** will use the system to track costing, process buyer payments, and manage supplier and employee payroll. **Production Incharges** (Supervisors) will rely on the platform to assign workers to production stages, monitor machinery usage, and generate Bill of Materials (BOM) requests. **Quality Control Inspectors** will interact with the system to log inspection results and clear final products for packaging. **Logistics Handlers** will use it to manage packaging and update shipment statuses. Finally, **System Administrators** will maintain the overall database integrity, manage user credentials, and ensure smooth system operation.

4. Front End and Back End Tools

- **Frontend:** PHP, Bootstrap (for responsive UI design), and Vanilla JS (for client-side interactions)
- **Backend:** PHP integrated with the OCI8 extension (for seamless Oracle database communication)
- **Database:** Oracle 11G (for handling complex schemas, vast operational data, and maintaining strict relational integrity)
- **Environment:** XAMPP / Apache Web Server

5. Features of Project

- **Order and Procurement Management:** Module to record incoming buyer orders, categorize required materials (Fabric/Accessories), and track material procurement from suppliers.
- **Dynamic Costing and BOM Calculation:** Automated calculation of order costs by aggregating Bill of Materials (BOM) requirements and machinery operational expenses.
- **Production and Floor Management:** Tracking of distinct production stages, machinery utilization, and mapping of specific workers and incharges to active manufacturing phases.
- **Quality Assurance Tracking:** Functionality to route production output for quality inspection and log the approval of final products.
- **Logistics and Shipping Control:** Management of packaging phases and tracking of final shipment dispatch to the buyer.
- **Integrated Financial Accounting:** Centralized ledger to handle incoming payments from buyers, outgoing payments to material suppliers, and comprehensive employee payroll disbursement.